

Disaster Management

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Abstract

Climate change is making natural disasters more frequent and severe, yet little is known about the capacity of firms to withstand such disasters and adapt to them. We examine these issues using the World Management Survey (WMS), which includes new questions on firms' climate change perceptions and climate adaptation behavior as well as management practices. Combining the WMS with geocoded data on natural disasters, we create a panel spanning almost 8,000 firms across 33 countries over three decades. Exposure to disasters is associated with lower firm survival and subsequent growth, but firms with structured management practices are more resilient, suffering smaller drops in jobs growth and lower exit risk. Exploring mechanisms, we show that firms with structured management practices have more accurate perceptions of climate-related risks to their businesses and are also more likely to have implemented measures to adapt to climate change. Our results are consistent with structured management practices contributing to firms' resilience to climate shocks, highlighting management as a policy-relevant margin for mitigating the harmful consequences of climate change.

JEL classification: Q54, M11, L25, H12

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1 Introduction

Climate change has made natural disasters more frequent and this trend is likely to continue over the coming decades (IPCC 2023). The impact of natural disasters on firm performance is therefore a growing economic and political concern. While a considerable body of research has documented that natural disasters tend to have negative economic impacts, little is known about how firm capabilities can ameliorate such negative effects. There is also a paucity of evidence on the extent to which firms are perceiving and adapting to the increased frequency of natural disasters and other risks induced by climate change.

In this paper we address both of these evidence gaps by examining the relationship between natural disasters, management practices and firm outcomes. We combine surveys on management practices (including a new wave of the World Management Survey, WMS), with performance data from digitized company accounts (BVD Historical Orbis) and natural disaster data from the Geocoded Disasters Dataset (Rosvold and Buhaug 2021). This novel data combination spans 33 countries over 35 years from 1984 to 2018.¹ Using this data, we show that exposure to natural disasters is associated with a higher probability of exit and lower subsequent growth. More interestingly, we find that firms with structured management practices appear to avoid some of these negative impacts with such firms experiencing significantly lower exit risk and smaller reductions in subsequent jobs growth immediately following a disaster. Looking over longer horizons, we find the negative effects of disasters and the offsetting effect of management attenuate to some degree over time, consistent with management primarily reducing short-run disruption.

Exploiting novel questions in the latest WMS wave, we examine the channels through which management practices may influence firms' resilience to climate-related risks. First, we examine *perception*-driven mechanisms and show that firms with more structured management are more likely to perceive climate change as a risk, especially when they are located in areas that have experienced increases in natural disasters over recent decades. Second, we study firms' *adaptive* responses and find that firms with more structured management are also more likely to implement measures to reduce climate-related risks, both overall and conditional on a given level of perceived risk. Together, these results are consistent with three mechanisms by which management can improve resilience to natural disasters. Management may improve how firms read climate-related signals in the places where those signals have changed, strengthen their capacity to act on any given signal, and bring organizational capabilities that raise adaptation irrespective of perceived risk. These mechanisms have distinct policy implications, and we discuss what our results imply for the effectiveness of climate adaptation policy.

The paper is structured as follows. We discuss the related literature in the rest of the introduction before describing the data we use to measure firm management practices, outcomes and their exposure to natural disasters in Section 2. Section 3 outlines our empirical strategy and presents results on the relationship between natural disasters and firm outcomes and the role

1. The panel dimension comes from the accounting and disaster data, which are available from 1984 and 1960 respectively, in comparison to the WMS data which is available from 2004. As discussed in Section 2, we treat management as a time-invariant firm characteristic, consistent with evidence that management practices are highly persistent, and therefore apply a firm's management score to all years in which we observe it.

of management in moderating these effects. Section 4 sets out the three mechanisms through which management may reduce the negative impact of disasters and provides evidence on each by examining associations between management, climate change risk perceptions and adaptation to such risks. Section 5 concludes.²

We contribute to at least three literatures. First, we relate to work examining the importance of management practices for economic performance. A large body of recent work has documented how management practices appear to be an important driver of firm-level and macro-level productivity.³ However, relatively little is known about *why* management matters so much. One under-explored mechanism is that structured management practices enable firms to be more resilient to negative shocks. Two papers examine this hypothesis using the COVID-19 pandemic as a large, negative shock. Lamorgese et al. (2024) exploit the timing of a survey of Italian firms that spanned the arrival of the pandemic, and find that a one standard deviation higher management score reduces the drop in year-ahead expected sales by 30 percent, with more structured firms more likely to reorganize labor and adopt remote working. Jones et al. (2026) study UK firms over 2019–2023 and find that those with more structured pre-pandemic practices experienced smaller falls in sales revenue, while adopting homeworking and online sales more fully and adapting supply chains more effectively. Both papers examine a single aggregate shock that reached all firms in their samples at essentially the same time, so the differential response to management is identified from how firms fared through one common event. We instead exploit many localized shocks occurring in different places at different times, which allows us to condition on expected exposure and to base identification on variation in the timing of disasters. Natural disasters are also more prevalent than pandemics, and increasingly so, making the question of which firms withstand them one of recurring rather than episodic relevance.⁴ Finally, our data allow us to observe firm exit, so we examine resilience on the extensive as well as on the intensive margin.

Another reason for the positive association between structured management practices and economic outcomes is that firms with such practices are more able to process and respond to relevant information. Supporting this hypothesis, we find firms with structured management are more likely to perceive climate-related risk and adopt measures to reduce their exposure to these risks. This is consistent with Bloom et al. (2025), who use a survey of over 20,000 UK firms that elicits both management practices and forecasts, and find that firms with more structured management practices forecast both their own sales revenue and aggregate GDP more accurately, as well as reporting lower subjective uncertainty around their central estimates. Bloom et al. (2025) document this association for forecasts of own-firm performance and macroeconomic conditions, which are outcomes central to firms’ core business operations. Our contribution is to show that the association extends to the perception of climate-related risk, a domain outside firms’ primary expertise. This suggests the link between management and informational

2. Online Appendices give more details on the data (Appendix A) and results (Appendix B).

3. See Scur et al. (2021) for a recent survey using the WMS. Other examples are provided by Bandiera et al. (2020), Bianchi and Giorcelli (2022), Bloom and Van Reenen (2007), Bloom et al. (2019), Bloom et al. (2020), Bruhn et al. (2018), Giorcelli (2019), Huber et al. (2021), Iacovone et al. (2022), and Lazear et al. (2015).

4. Aghion et al. (2021) find that more decentralized firms performed better throughout the negative shocks of the financial crisis. We show that decentralization and other potential correlates of management do not drive our results.

accuracy reflects a general organizational capacity rather than domain-specific business knowledge. It also provides empirical evidence that structured management practices can contribute to firms’ ‘dynamic capabilities’ (Teece 2007), via an improved awareness of environmental risks. Some papers emphasize that managerial dynamism can improve business performance through greater appreciation of economic factors such as market trends (Augier and Teece 2009); we find this also applies to awareness of factors in the natural (rather than business) world, which can threaten firms’ operations.

Some existing papers examine links between management and climate change mainly focusing on climate change mitigation (i.e. actions to reduce firms’ emissions). An early example finding a positive association between climate mitigation and management practices was Bloom et al. (2010), who study over 300 UK manufacturing firms documenting that those with more structured management were not only more productive but also less energy and carbon intensive. Martin et al. (2012) find similar patterns using a measure of specifically green (rather than general) management practices, whereas Capelle et al. (2024) provide evidence from a global sample of publicly listed firms that ‘laggards’ with high energy intensity relative to others in their industry and country exhibit relatively poor management practices. An interpretation of these correlations is that managers may be better informed about the costs and benefits of energy efficiency improvements and suffer less from present-bias (Allcott et al. 2014). Haas et al. (2022) provide direct evidence of this channel by drawing on World Bank surveys in emerging economies to show firms with better green management practices invest more in green technologies. Shin (2023) provides evidence of another mechanism between management practices and energy use, showing that firms with structured management practices are more responsive to increases in fuel prices.

Far fewer authors have examined links between management and adaptation to environmental risks (i.e. actions taken by firms to reduce the impact of negative environmental shocks on their performance), which is the focus of this paper. Indeed, the only work on this subject that we are aware of is Adhvaryu et al. (2022), which shows how management can reduce the negative impact of pollution shocks on productivity. Using detailed data on productivity, task assignments, and managerial characteristics within a large Indian garment firm, they find that air pollution shocks negatively impact worker-task productivity but that managers are able to mitigate the impact of such shocks by reallocating negatively-affected workers to less pollution-sensitive tasks. Importantly, they document significant heterogeneity in the efficacy of such mitigation according to manager attentiveness: a one standard-deviation increase in manager attentiveness almost entirely offsets the negative productivity impact of pollution experienced by the average manager. Despite the specificity of the setting and their focus on productivity at the granular worker-task level, the findings in Adhvaryu et al. (2022) are broadly consistent with our own to the extent that manager attentiveness can be interpreted as an example of more structured management practices.

A second substantial body of related work examines the economic impacts of extreme weather and natural disasters, spanning analyses at the country level (Dell et al. 2012; Acevedo et al. 2020), the local level (Kocornik-Mina et al. 2020; Elliott et al. 2015), and the firm level. Within this last strand, a growing literature documents significant productivity losses from cli-

matic shocks. Zhang et al. (2018) use detailed production data on Chinese manufacturing firms to establish an inverted-U relationship between temperature and productivity, with high temperatures generating particularly large negative effects. Somanathan et al. (2021) similarly find that a one-degree Celsius temperature rise reduces annual output among Indian manufacturing plants by approximately 2%, driven primarily by the effect of heat on labor productivity. Extending the focus beyond temperature, Clò et al. (2024) show that exposure to floods and landslides increases firm exit and reduces sales and employment among Italian firms, with effects concentrated among small and low-tech establishments. Erda (2026) focuses on the US and finds that low productivity firms exit after large floods leading to a subsequent positive effect on aggregate productivity via capital upgrading and reallocation facilitated by federal disaster spending. Our paper contributes to this literature by identifying structured management practices as a key source of heterogeneity in disaster impacts across firms. In particular, we show that firms with more structured management practices (which are typically more productive) are significantly less likely to exit following natural disasters, a finding that complements the dynamics documented by Erda (2026).

Several papers extend the analysis of firm-level disaster impacts beyond direct effects, considering how inter-firm networks cause shocks to propagate. Barrot and Sauvagnat (2016) find that natural disasters in the US that affect supplier firms in production networks also cause losses in output and market value among downstream consumer firms. The impacts they find are particularly pronounced if the impacted supplier produces relatively specific inputs making it difficult for downstream consumers to substitute to other, non-affected firms. Tanaka (2015) and Carvalho et al. (2021) both examine how the impact of the 2011 Great East Japan Earthquake and subsequent tsunami was amplified via firm networks. Carvalho et al. (2021) focus on domestic firm networks and document transmission of negative disaster impacts both up and down supply chains while Tanaka (2015) shows the impact of the event extended beyond Japan itself with negative impacts on the output of US branches of Japanese multinationals in proportion with the reliance of the US firms on inputs from Japan. In contrast to these papers, we focus on the impacts that extreme environmental events have on firms that are themselves exposed to such disasters.⁵ Given the consistent evidence of indirect impacts provided by these papers, our omission of such propagation effects means our findings should not be interpreted as estimates of the comprehensive economic impact of disasters. Rather, our focus is on documenting whether heterogeneity in the effects of disaster exposure can be attributed to differences in management practices, which we indeed find to be associated with differences in the direct negative firm-level impacts we estimate.

Third, we contribute to an expanding literature related to climate change adaptation. A large proportion of work in this area focuses on agricultural adaptation, highlighting the capacity for insurance and novel technologies to reduce the negative effects of environmental shocks (for example see Dar et al. (2013), Auffhammer and Carleton (2018), Michler et al. (2019) and Hultgren et al. (2022)). Empirical studies of adaptation among non-agricultural firms are less

5. As explained in Section 2, our measure of ‘exposure’ is defined as being located within a 40km radius of a disaster’s centre. Therefore, while we do not explicitly aim to measure indirect impacts of disasters that arise through supply network linkages, the effects we estimate may implicitly capture localized indirect impacts due to suppliers that are located close to the firm.

numerous but have documented several changes in firm practices that can mitigate the adverse effect of climate shocks such as the adoption of LED lights (Adhvaryu et al. 2020), worker responses such as timing of breaks (Masuda et al. 2021), and changes in supplier networks (Balboni et al. 2023). In relation to this literature, our results suggest management practices affect a firm’s ability to implement such adaptive changes.

2 Data

Our aim is to examine the relationship between firm performance, management practices and natural disasters. In this section we describe the data we use to measure each of these factors and document the characteristics of our analysis sample. More details are provided in Appendix A.

2.1 Firm Outcomes and Management Practices

Given our focus on the relationship between firms’ management practices and the impacts of natural disasters, we take data on firms from the latest version of the World Management Survey (WMS). The WMS is an interview-based survey conducted by trained analysts who engage middle managers in semi-structured conversations about daily practices within their respective establishments, as outlined by Bloom and Van Reenen (2007). The interview format is designed to prevent interviewees from providing responses they believe the interviewer wants to hear and covers a range of practices related to operations, monitoring, target setting, and people management systems. Within each of these areas, there are 18 topics, with each topic assessed on a scale from 1 (indicating little to no structure or weak practices) to 5 (indicating well-structured or best practices). Pooling across all WMS survey waves results in a dataset of 17,783 observations of 13,721 manufacturing firms.

We construct indices of management from the individual WMS scores by z-scoring each question, averaging across all management questions within a firm and then z-scoring again the resulting measure to obtain an overall management score with mean zero and standard deviation 1. Similar management indices constructed from the WMS data have been shown to have significant predictive power of a range of dimensions of performance, with more structured firms having higher productivity, profitability and growth rates (Scur et al. 2021).

In the second part of our empirical analysis we draw on two survey measures that speak to the mechanisms through which management may mediate the impact of natural disasters on firms, namely firms’ awareness of climate-related risks and their adoption of measures to mitigate such risks. To explore these issues, we leverage the following questions in the latest wave of the WMS, which was conducted over 2021, 2022 and 2023:

1. *Climate Risk Perception* “Would you say things like rising temperatures, natural disasters, and changing seasonal weather patterns can put the operation of your manufacturing site at risk?”
2. *Climate Adaptation* “Do you have any measures in place that are responses to the potential effects of climate change? Can you tell me about some of these measures?”

As with the management scores these are on a 1 to 5 scale, which we standardize. Examples of actions firms report under the climate adaptation question are provided in Figure B2 and Table B23 in Appendix B.

Two issues arise over the climate-related questions. First, the wording of the climate risk perception question means a firm could answer ‘no’ if they did not regard climate change as a material threat to their operations or if they did regard climate change as a risk but believed the actions they had taken were sufficient to protect their manufacturing site. However, the discursive nature of the WMS survey process means that the initial response is followed by subsequent ‘why’ questions. The eventual score against this question is given based on an assessment of the initial and follow-up answers and will therefore reflect any evidence that firms have considered temperature, disaster or weather-related risks as factors that could compromise their operation. Second, the wording of the perception question is relatively suggestive compared to the management practice questions in the WMS, in that it explicitly highlights a number of distinct examples of climate-related risk. While this may push respondents toward a ‘yes’ answer to some extent, in practice around a third of companies are scored 1-2 against this question, indicating little awareness whereas only 16% are given a score of 5, indicating thorough and considered awareness.

To measure firms’ financial outcomes we combine the WMS with Historical Orbis (HO), a database of essentially all the registered firms in the countries in our data. HO has digitized versions of their company accounts enabling the construction of a panel dataset containing employment, capital (fixed assets), value added and survival. Changes in company identifiers mean we are unable to match all WMS firms with an HO record. Furthermore, since accounting standards differ over what information must be disclosed between countries, firm size classes and time periods, not all WMS firms have full information on inputs and outputs.⁶ We are able to obtain some financial information for 11,629 of the 13,721 WMS firms resulting in a panel of 175,525 firm-year observations.

We treat management as a time-invariant firm characteristic in our analysis, measured as the average of a firm’s structured management score across all WMS waves in which it appears. This approach is consistent with prior evidence that structured management practices are highly persistent (e.g. Bloom et al. (2019)). We also control for workforce skills using the share of employees with a college degree recorded in the WMS. Since workforce composition exhibits more time-series variation than management practices, we linearly interpolate this measure between waves for firms observed more than once (results are robust to dropping the skills control entirely; see Appendix Table B17).

2.2 Natural Disasters and Disaster Exposure

Data on natural disasters are taken from the Geocoded Disasters Dataset (GDIS) (Rosvold and Buhaug 2021). GDIS contains geocoded information on natural disasters recorded in the

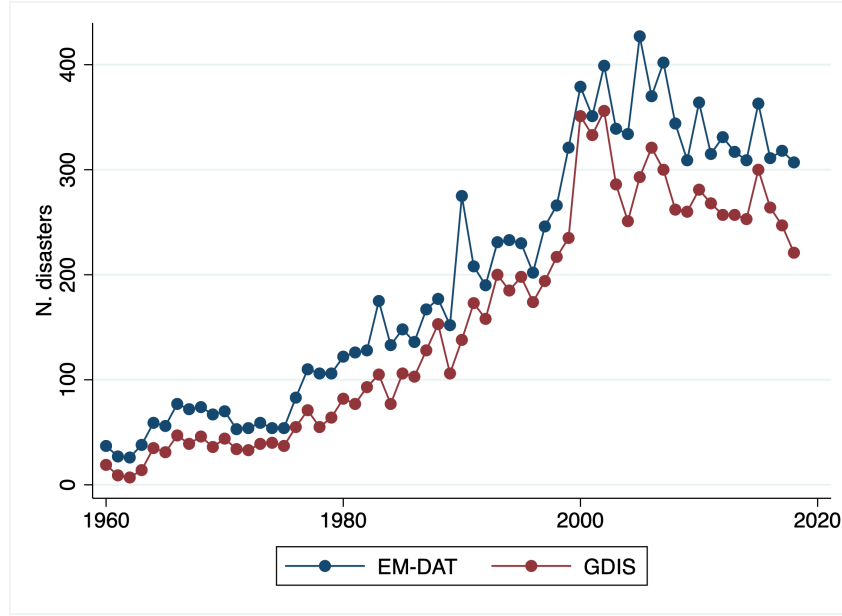
6. Firm information recorded in HO is sometimes at a higher level than in the WMS, for example if the entity surveyed in WMS is a plant of a larger company. The HO database also frequently contains multiple records for a given company-year observation if, for example, companies publish financial reports in both annual reports and local registry filings. We proceed by taking mean financial variables within an HO firm-year combination and combining with the WMS data at the lowest level possible.

Centre for Research on the Epidemiology of Disasters' Emergency Events Database (EM-DAT), between 1960 and 2018. The EM-DAT database is a worldwide record of complex emergencies with natural disaster information covering floods, storms, earthquakes, volcanic activity, extreme temperatures, landslides, droughts, and (dry) mass movements. An event must meet specific criteria related to its impact to be included in the EM-DAT database. Generally, an event is considered for inclusion if it results in ten or more deaths, 100 or more people affected, a declaration of a state of emergency, or a call for international assistance. Combining information from various sources, including government reports, international organizations and news agencies, the database is regarded as a relatively comprehensive record of natural disasters and has been used in several previous studies of their effects including Kahn (2005), Besley and Persson (2009) and Cavallo et al. (2013).

EM-DAT provides standardized information on events' date, type, impact, and the number of casualties, but lacks standardized measures of disaster location. Locations are recorded as the names of affected places with no standardized unit and sometimes are recorded as geographical features, such as a mountain range, rather than a particular settlement. GDIS addresses this by matching the location names in EM-DAT to standardized units taken from the Global Administrative Areas database (GADM 2018) and obtains latitude and longitude coordinates of the unit centroids. GDIS only includes disasters located with what the data creators deemed to be an adequate degree of accuracy. In practice, 85% disaster-locations in EM-DAT were successfully geocoded with floods and earthquakes achieving nearly-100% and droughts the lowest coverage at 64%. Since the majority of disasters in EM-DAT are floods and storms, the two datasets track each other closely (see Figure 1).⁷

7. Appendix Figures A1 and A2 show the number of disasters by type and country respectively. Further details on the algorithm used by the creators of GDIS to obtain disaster locations is provided in Rosvold and Buhaug (2021). In 2014 EM-DAT enhanced its location data by geocoding natural disaster records, with the exception of biological disasters, starting from the year 2000 onwards. Geocoding was conducted by matching textual information on the affected area to Global Administrative Unit Layers (FAO) area codes. GDIS geocoded EM-DAT disasters using similar approach but starts in 1960 rather than 2000 and, in some cases, contains geographic data at a finer spatial resolution than the geocoded EM-DAT. We use GDIS rather than the geocoded version of EM-DAT for these reasons.

Figure 1: Natural Disasters Recorded in EM-DAT and GDIS



To measure firms' exposure to natural disasters, we first geolocate firms in the WMS by searching Google Maps for their name, country and, if present, city and zipcode using the Google Maps Geocoding API. The API call returns the coordinates of the location that best matches the input string, although fails if no match or only an imprecise match is found. Of 13,721 WMS firms we are able to accurately geolocate 10,154.⁸ We then calculate the linear distance between firms' precise location and the location centroid for every natural disaster recorded in GDIS and classify firms as exposed to any particular event if this linear distance is less than or equal to a threshold distance.

Several considerations influence the choice of the threshold distance used to define disaster exposure. While a narrow threshold confines the disaster exposure definition to firms that are proximate to, and thus likely more affected by, disasters, it also reduces the number of 'disaster exposed' observations. Narrow thresholds may also miss spillover effects to firms located in areas surrounding those affected by a disaster, such as changes in local demand and labor markets. Absent any 'correct' way to balance these considerations, we estimate our main results using various disaster exposure thresholds ranging from 20km to 70km. Results presented in the main body of this paper are based on classifying a firm as exposed to a particular event using a 40km threshold, since results obtained using this definition are representative (in terms of sign and significance) of those obtained using both narrower and wider thresholds. We present additional estimates of our main specifications using the full range of thresholds in Appendix Figure B1. Using the 40km-based definition of disaster exposure, we take the count of disasters a firm is exposed to in a given year and construct a binary measure of disaster exposure if this sum is greater than 0.

While the location information provided in GDIS is a considerable improvement on that available in EM-DAT, there are two significant reasons why it remains an imprecise record of

8. To improve accuracy, we disregard geocode results that pertain to the centroid of an administrative area e.g. a state/country.

natural disaster locations. First, the algorithm used to geolocate EM-DAT disasters failed in some instances where the EM-DAT location was either very general, such as a river running through large parts of a country, or non-unique within a country. Second, the specific latitudes and longitudes assigned to a disaster pertain to the centroids of the affected administrative units rather than boundaries that demarcate the extent of the disaster and are therefore approximations of the true boundaries of affected areas. Since a single disaster can be associated with multiple administrative units, the precision of this approximation is inversely related to the granularity of the administrative units: for example, it is safer to assume the entirety of a unit was flooded by an event ascribed to the centre of the unit when the unit is a village rather than a state. Although the highest spatial resolution in GDIS corresponds to administrative level 3 (usually district), the vast majority of the locations are administrative level 1 (typically state/province). Such imprecision of GDIS event locations therefore introduces a degree of measurement error into our measures of firms' exposure to disasters: our measure will omit disasters that GDIS failed to geocode and may incorrectly classify a firm as exposed if the area truly impacted by the disaster is further away than that implied by a GDIS centroid.

Appendix B.1 formally examines the implications of disaster exposure misclassification for our analysis. If misclassification is not systematically related to management, measurement error attenuates estimates of the impact of disasters on firms and the extent to which these impacts are mitigated by management practices. More generally, if misclassification varies systematically with management, estimated coefficients may in principle be biased in either direction. To assess whether differential misclassification is a concern in our setting, we compare floods and storms recorded in GDIS to a distinct flood dataset, which relies more heavily on satellite-derived measures of inundation. While alignment between the two sources is not perfect, the results reported in Appendix B.1 provide no evidence that misclassification is systematically related to management practices. This suggests that any misclassification of disaster exposure in our data is more likely to attenuate estimated effects than to generate misleading results. The fact that we observe significant impacts of disasters on firms despite such attenuation bias reassures us that our measure of disaster exposure, while inevitably imperfect, retains adequate signal to provide insight.

2.3 Sample Characteristics

Table A2 presents sample means for several firm characteristics. Our analysis sample is restricted to firms in the WMS that were successfully geolocated and matched to GDIS and accounting data. This gives us an analysis sample of 7,899 firms and 112,783 observations in 33 countries between 1984 and 2018. The sample begins in 1984, as this is the first year of HO accounting data, and ends in 2018, as this is the last year of the GDIS disaster data.

The WMS has a sampling frame of manufacturing firms with between 50 to 5,000 employees, so results may not be generalizable to very small firms (where there may be less need for formal structured management practices) and very large firms.⁹ Previous work shows that compared to this frame of medium sized manufacturing firms, WMS respondents are broadly balanced on

9. Table A1 in Appendix A shows the country composition of our analysis sample.

observable characteristics such as size, productivity and industry.¹⁰ Table A2 shows that relative to the full WMS sample, our firms are more European, predominantly due to higher coverage in the HO data. They are also larger with slightly higher management scores (0.01 of a standard deviation larger than the full sample).

3 The Impact of Natural Disasters on Firm Performance

3.1 Econometric Strategy

In this section we examine the impact of natural disasters on firm performance and whether such impacts are offset by structured management practices. First, we focus on the impact of natural disasters ignoring any heterogeneity in their effects. Our empirical strategy exploits variation in the timing of natural disasters. As emphasized by Borusyak and Hull (2023), research designs that rely on plausibly exogenous shocks can be biased when units exhibit predictable variation in their exposure to those shocks. In our setting, predictable exposure arises if certain types of firms disproportionately locate in areas with persistently high disaster risk, for example due to historical specialization or offsetting locational advantages such as market access. Moreover, because disaster incidence is serially correlated, firms observed in high-risk areas may differ systematically from other firms as a result of past disaster exposure, potentially inducing correlation between disaster exposure and firm performance.

To address these concerns, we follow the recommendation of Borusyak and Hull (2023) to model and control for expected shock exposure. Specifically, we estimate expected disaster occurrence, $\mathbb{E}[D]_{it}$, as the fitted value from a regression of the form:

$$D_{it} = \pi_c D_{iPRE10} + \phi_{rt} + e_{it}, \quad (1)$$

where D_{it} is disaster exposure as discussed in Section 2, namely whether there was a disaster in year t within 40km of firm i . D_{iPRE10} is a set of dummies for the number of disasters within 40km of firm i in the decade prior to the first time we observe the firm and we allow these effects to differ by country (by allowing for country-specific parameters, π_c). The ϕ_{rt} are region-by-year dummies, which allow the trend in disaster exposure to vary across six broad geographic regions (Europe, North America, Central and South America, Africa, Asia and Oceania). Finally, e_{it} is an error term. This specification captures persistent, location-specific disaster exposure, which we then control for in regressions that relate disaster occurrence to firm performance.¹¹

By conditioning on expected disaster exposure, identification of disaster effects is therefore based on residual variation in disaster timing conditional on underlying risk. Under the assumption that such residual variation is orthogonal to firms' performance trajectories, the effect of

10. For example, see Bloom and Van Reenen (2007), Bloom et al. (2014) and Bloom et al. (2024).

11. Our main results are robust to many alternative specifications of the 'expected disaster' equation (1), such as including country-by-year fixed effects or just year fixed effects instead of region-by-year fixed effects; using a five-year 'pre' horizon rather than a ten-year horizon or imposing common coefficients on D_{iPRE10} across countries. Our findings also hold when using just D_{iPRE10} instead of $\mathbb{E}[D]_{it}$ in equations (2) and (3), or even dropping $\mathbb{E}[D]_{it}$ completely. Sample averages of the D_i and D_{iPRE10} variables are given in Appendix Table A2.

disaster exposure can be estimated using:

$$\Delta y_{ijct+1} = \beta_0 D_{it} + \beta_1 M_i + \beta_2 X_{it} + \beta_3 \mathbb{E}[D]_{it} + \tau_t + \iota_j + \kappa_c + \epsilon_{it}. \quad (2)$$

Δy_{ijct+1} is the growth (between $t+1$ and t) of a particular outcome such as inputs, outputs or is a dummy for survival. M_i is our WMS measure of structured management practices, X is a vector of firm characteristics including firm age, the share of employees with a degree and firm size. To avoid any mechanical correlation between outcomes and controls, firm size is measured as log fixed assets when the outcome variable is employment growth and as log employment otherwise. τ_t , ι_j and κ_c are vectors of time, two-digit industry and country of location dummies respectively and ϵ_{it} the error term (we cluster standard errors by firm to allow within-firm correlation).

We examine whether management practices affect the impact of natural disasters using generalizations of equation (2) of the following form:

$$\Delta y_{ijct+1} = \beta_0 D_{it} + \beta_1 M_i + \beta_2 X_{it} + \beta_3 \mathbb{E}[D]_{it} + \beta_4 (D_{it} * M_i) + \beta_5 (D_{it} * X_{it}) + \tau_t + \iota_j + \kappa_c + \epsilon_{it}. \quad (3)$$

This specification allows the effect of disaster exposure to vary with firm characteristics. The coefficient of primary interest is β_4 , which captures whether the impact of natural disasters differs systematically with structured management practices. We also include interactions between disasters and other firm attributes, X_{it} , to accommodate heterogeneity in disaster impacts along other observable dimensions, and de-mean any variable interacted with the disaster indicator so that the β_0 coefficient can be interpreted as the impact of disaster on a firm with average characteristics. In addition, we consider a more parsimonious variant of equation (3) that excludes the $(D_{it} * X_{it})$ terms.

The Δy_{ijct+1} we consider include survival, a dummy that equals one if the firm survives between t and $t+1$ (or longer windows in some specifications) and zero otherwise (i.e., it exits). Aside from survival, all outcomes we consider are *changes* in firm-level metrics such as capital, employment and value added. Specifically,

$$\Delta y_{ijct+s} = \frac{\ln(y_{it+s}) - \ln(y_{it})}{s}, \quad (4)$$

where y is capital, employment, value added. Our baseline results use the one-period growth rate, $s = 1$, but we also consider higher values of s to examine whether impacts are short-lived or persist over several years and normalize by s , to annualize growth rates and thus make them comparable across different time horizons. By focusing on growth rates, any constant differences across firms in the levels of outcomes are differenced out. We trim growth rates at the top and bottom 2% level to reduce the impact of outliers.

Several points regarding equation (2) warrant further discussion. First, while the specification echoes those of conventional event-studies, it includes a single disaster indicator rather than pre- and post-event dummies or distributed leads and lags. We adopt this approach because defining clean dynamic profiles is complicated when firms experience multiple disasters within short time windows, which, although uncommon, does occur in our data. Appendix B presents complementary evidence from distributed-lag models following Dell et al. (2012). These results

are consistent with the dynamic patterns obtained from equation (3) and are discussed further alongside our main results.

A second issue concerns whether the coefficients on disasters, D , and its interactions can be interpreted as capturing the causal impact of disasters on firm performance. As discussed above, our principal identification assumption is that the timing of disasters is exogenous to firm performance conditional on controls. Beyond the concern over disaster predictability addressed above, a distinct concern is reverse causality. Disaster exposure could itself affect management practices, in which case the interaction term ($D_{it} * M_i$) might reflect endogenous changes in management rather than heterogeneity in disaster impacts. This would be especially problematic if the firms that change their management practices following disaster exposure were on different growth trajectories to firms that were exposed to disasters but did not change their management practices. In Appendix B.2 we analyze a subsample of firms that were surveyed at least twice by the WMS to directly test whether disaster exposure predicts changes in management practices and whether any effect differs according to a firm’s pre-disaster growth trajectory. Across specifications, we find management practices are highly persistent and that there is no systematic evidence of disaster exposure affecting changes in management practices. These results mitigate concerns of reverse causality and support the interpretation of the interaction coefficients in equation (3) as capturing heterogeneity in the response of firm performance to disasters.

Finally, the strategic location of firms could, in principle, generate correlation between more management practices and disaster exposure if firms with more structured management systematically sort into locations with lower (or higher) disaster risk. Such sorting would bias the estimated disaster and disaster–management interaction effects by conflating the impact of disasters with pre-existing differences in firms’ location choices and growth trajectories. However, Appendix Table A3 shows no significant association between management practices and disaster exposure once we condition on the firm characteristics included in our main specifications.

Taken together, these considerations delimit what our estimates can establish. Our identifying assumption is that, conditional on expected exposure and the controls described above, the timing of disasters is orthogonal to firms’ performance trajectories. The evidence discussed in this section is consistent with that assumption but cannot verify it, and management practices are of course not randomly assigned across firms. For readability we use terms such as ‘effect’, ‘impact’ and ‘resilience’ throughout, but these should be understood as shorthand for the associations we estimate under this assumption rather than as claims of established causality.

3.2 Main Results

Table 1 shows regression estimates where the dependent variable is capital growth in column (1), employment growth in column (2), value added growth in column (3) and survival in column (4). Different panels present variants of equation (3), with year and country dummies in all panels. There are 7,899 firms in total, but the exact number differs across columns due to missing values on the outcome variables, largely reflecting different accounting regulations.

Table 1: Impact of disaster exposure and management on one year ahead firm performance

	(1)	(2)	(3)	(4)
	Log differences ($t, t + 1$)			
	Capital	Emp.	VA	Survival
Panel A: Linear Management				
Management	0.0024** (0.0011)	0.0018** (0.0008)	0.0041*** (0.0013)	0.0181*** (0.0047)
Panel B: Linear Disaster Exposure				
Disaster in year	-0.0077*** (0.0023)	-0.0031** (0.0014)	-0.0087*** (0.0030)	-0.0326*** (0.0114)
Panel C: Linear Firm Characteristics				
Disaster in year	-0.0071*** (0.0023)	-0.0025* (0.0014)	-0.0083*** (0.0030)	-0.0369*** (0.0114)
Management	0.0016 (0.0011)	0.0017** (0.0008)	0.0064*** (0.0014)	0.0090* (0.0050)
Panel D: Linear Firm Characteristics and Management-Disaster interaction				
Disaster in year	-0.0071*** (0.0023)	-0.0024* (0.0014)	-0.0083*** (0.0030)	-0.0366*** (0.0113)
Management	0.0014 (0.0012)	0.0010 (0.0009)	0.0056*** (0.0015)	0.0039 (0.0054)
(Management)* (Disaster)	0.0010 (0.0023)	0.0038*** (0.0014)	0.0043 (0.0031)	0.0282** (0.0120)
Panel E: Linear Firm Characteristics and Full Disaster interactions				
Disaster in year	-0.0070*** (0.0023)	-0.0022 (0.0014)	-0.0082*** (0.0030)	-0.0379*** (0.0114)
Management	0.0014 (0.0012)	0.0012 (0.0009)	0.0055*** (0.0015)	0.0039 (0.0054)
(Management)* (Disaster)	0.0008 (0.0024)	0.0027* (0.0015)	0.0044 (0.0033)	0.0286** (0.0125)
Dep. var. mean	0.026	0.006	0.033	0.982
N obs.	73869	59487	33700	112783
N firms	5288	4706	2580	7899

Note: standard errors in parentheses clustered at the firm level. Differences in sample size across columns are due to missing values in the HO financial data. Dependent variable in columns (1)-(3) are one-year differences in the log of the outcome denoted in the column title defined according to equation (4). Dependent variable in column (4) is a dummy equal to 1 if the firm survived to year $t + 1$ and zero if it exited. All columns include country and time dummies. Panels C, D and E also include two digit industry dummies. Firm characteristics are firm age, size and skills. Coefficients and standard errors in column (4) are multiplied by 10. */**/** denote significance at the 10/5/1 percent level respectively.

Panel A of Table 1 begins with management correlations and shows that firms with higher management scores have significantly faster growth in inputs (first two columns), output (third column) and higher survival (final column). Panel B has the same specification, but uses disasters instead of management on the right hand side. This validates the disaster variable, showing that exposure to natural disasters is associated with significantly lower growth in firms' capital, employment and value added and with a lower survival probability. The estimates of these effects are economically important when compared to the mean value of the outcome variables (shown

at the base of each column). The impacts on growth in capital, employment and value added amount to 30%, 52% and 26% reductions relative to the mean observed growth rates, while the impact on firm survival corresponds to roughly an 18% increase in the firm exit rate.¹² Panel C has management and disaster in the same regressions and includes additional linear controls, showing a broadly similar pattern of coefficients.

Panel D of Table 1 is a key result, which generalizes the specification in Panel C to include the interaction between disasters and management from equation (3). Panel E additionally includes interactions between other firm characteristics (firm age, size and the proportion of employees with college degrees) and disaster exposure. In both panels, column (4) reveals that firms with higher structured management practices have a significantly higher survival probability following disaster exposure. A firm with a one standard deviation higher management score is associated with an offset of about three quarters of the impact of disaster exposure (e.g., in Panel D, $0.0282/0.0366 = 0.77$).

Even conditional on survival, jobs growth is significantly worse in disaster areas for firms with less structured management practices in column (2) of Table 1. The magnitude of the interaction coefficient is roughly equal to the linear disaster effect indicating that firms with one standard-deviation higher management scores experience essentially no negative impacts of disaster exposure on jobs growth.¹³ While the coefficients in the survival regression in column (4) were very similar in Panels D and E, the introduction of other disaster interactions causes the coefficient on the management-disaster interaction on employment growth to decline somewhat (from 0.0038 to 0.0027), suggesting that part of the estimated association reflects differences in other firm characteristics.

The coefficient on the interaction of management and disasters is also positive for capital growth and value added growth in columns (1) and (3), although these are not significant at conventional levels. Table B3 in Appendix B.3 shows estimates where the dependent variable is profits growth and total factor productivity growth. There is some evidence that disasters have a negative effect on these outcomes, although the estimates are imprecise. Similarly, the coefficients on the management-disaster interaction are insignificant.

What could explain this difference across outcomes? A first point to note is that we see similar results in columns (1) to (3) when imposing a common sample across outcomes, indicating that the pattern is not merely reflecting differences in sample composition (see Appendix Table B4).¹⁴ One explanation for the weaker effects of management-disaster interaction in the capital

12. Recall that our firms are in the manufacturing sector, where growth is relatively slow.

13. If we interpret these as the unbiased causal estimates, we can calculate the overall impact of management from extensive and intensive margins together. Consider two industries that are identical except the ‘high management industry’ has management scores one standard deviation larger than the ‘low management industry’. If both industries are simultaneously hit by a disaster and responses are given by Panel E, industry jobs initially fall half a percent (0.56%) faster in the low management industry. This splits about 50-50 along the intensive (0.0027) and extensive margins (0.0029).

14. We cannot estimate the column (4) survival results on a common *observation* sample as, by definition, all firms for whom we can estimate one-year-ahead growth survive for at least one year. When we restrict analysis to a common set of *firms*, as shown in Appendix Table B5, the management–disaster interaction on survival remains positive but becomes insignificant. This is because low-income countries account for 24% of the full sample but only 1% of this common-firm sample. As discussed in Section 3.4, the survival interaction is driven by firms in developing countries where external institutional buffers are weakest and the survival result should therefore be interpreted as applying primarily in such settings.

growth regression relates to adjustment costs. These are well-known to be much larger for capital than for labor. Below, we examine such dynamics explicitly using various techniques such as distributed-lag models. Appendix Table B6 for example, confirms an insignificant contemporaneous management–disaster interaction on capital growth, but a *significant* positive interaction at a one-year lag.

For the value added growth equation, the magnitude of the management interaction term is much larger than for capital (0.0044 vs. 0.0008 in Panel E) and more comparable to that on employment. From a production function perspective, since value added depends on labor inputs (where we find immediate effects of management on offsetting disasters) and capital inputs (where we do not), the weaker effects are unsurprising. Moreover, there is more measurement error in value added than in employment, so this contributes to the imprecision.

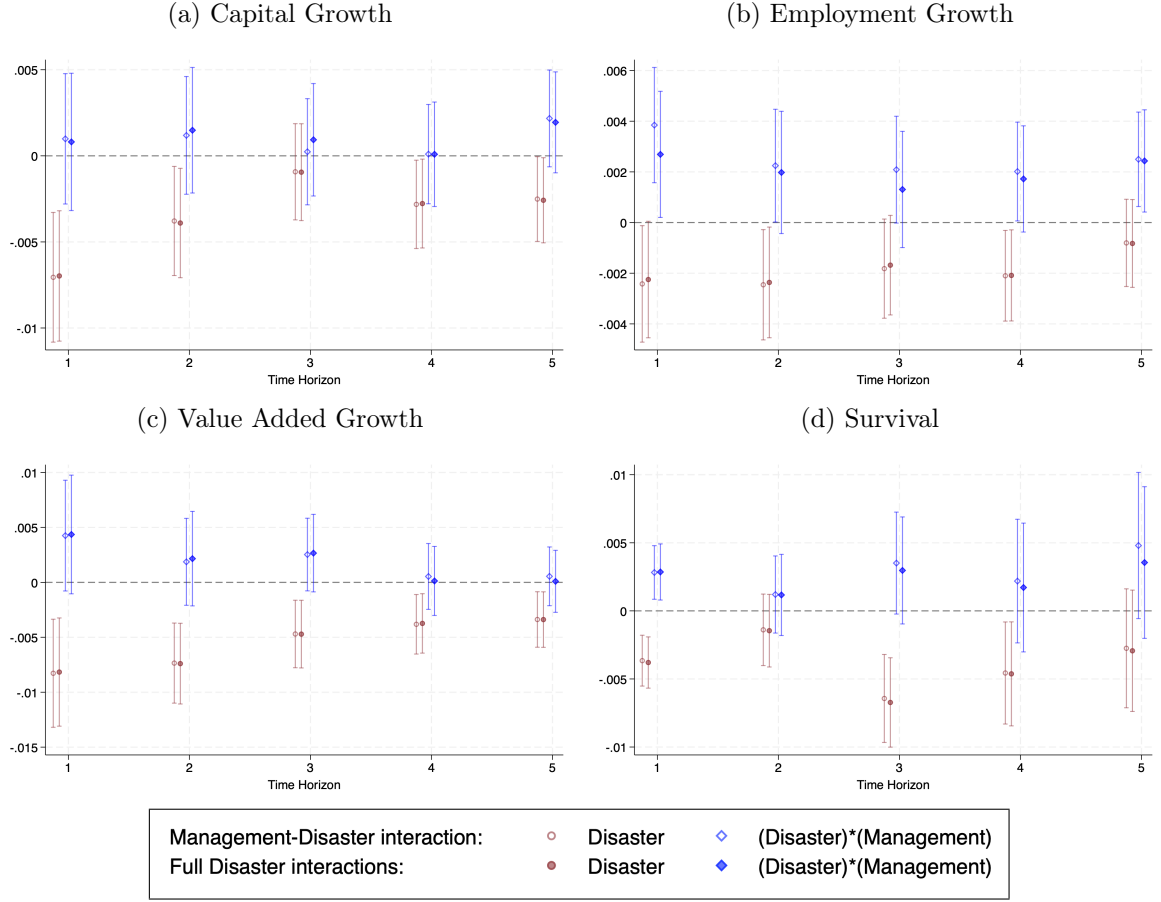
Overall, these patterns suggest that management’s protective association is clearest on the margins where managers can act quickly – retaining workers and keeping the firm operational – with effects on capital and output that are positive but slower to materialize and harder to detect in our data. This interpretation is further supported by the absence of significant management–disaster interactions on TFP and profits, which suggests that structured management does not substantially alter the impact of disasters on firm efficiency. Instead, the combined evidence points to management practices affecting firms’ capacity to maintain the scale of their operations through a disruption. This could, for example, be due to structured management practices helping firms to sustain liquidity, cover fixed costs or retain key workers. While it is challenging to examine these hypotheses in our data, identifying the specific mechanisms through which structured management practices protect firms from disasters remains an important question for future work, an issue that we return to in the conclusion.

3.3 Dynamic Effects

The analysis so far focuses on one-year ahead impacts. To examine the dynamic effects of disaster exposure, and the extent to which they depend on management practices, we re-estimate equation (3) examining outcomes 2, 3, 4 and 5 years after a disaster occurs. This approach is analogous to the local projections method of Jordá (2005), which has become a standard tool for estimating impulse responses in panel settings. Each panel of Figure 2 displays the estimates and 90% confidence intervals of equation (3), reflecting the linear impact of disasters (β_0) and the extent to which this is mediated by structured management practices (β_4). The x-axis is the time horizon where ‘1’ replicates Panels D (unshaded) and E (shaded) of Table 1.

Several patterns emerge from Figure 2. First, looking at the direct negative effects in red circles in the lower part of each panel (β_0), we see that the negative impacts on inputs and output growth tend to reduce over time, especially for value added. The pattern for survival is noisier, with standard errors becoming wider at longer horizons. Second, turning to the offsetting effect of management, we see that the disaster–management interactions in blue diamonds (β_4) dissipate over time for value added growth.

Figure 2: Main Parameter Estimates Over Different Time Horizons



Note: figure shows estimates of the β_0 and β_4 parameters from equation (3) obtained when outcomes defined over various time horizons, indicated by the x-axis labels. Maroon circles represent the β_0 disaster effect, while blue diamonds represent the β_4 coefficient on the interaction of management practices with disaster exposure. The hollow markers that are slightly offset to the left show estimates from a specification where only management is interacted with disaster exposure (equivalent to Panel D of Table 1), while the solid markers slightly offset to the right show estimates from a specification where all firm characteristics are interacted with disaster exposure (equivalent to Panel E of Table 1). The dependent variable in Panels (a) to (c) is the x-year annualized growth rate of the variable in the subfigure title, where x is the number of years between t and $t + x$. In Panel (d) the outcome is a binary indicator for survival by at least year $t + x$. Bars indicate 90% confidence intervals.

Appendix Tables B6 to B9 examine dynamic impacts using distributed-lag models similar to those used by Dell et al. (2012) to examine the impact of temperature shocks. In line with the results above, the negative impact of disasters are generally strongest in the year or two after impact. The offsetting effect of management for employment growth and survival is also strongest in the first year, with no significant evidence of an effect at longer lags. For capital growth, Table B6 shows little impact of the management interaction in the year of disaster exposure, but a strong positive and significant coefficient on the interaction with a one-year lag. This suggests that firms with more structured management suffer less pronounced slow downs in capital growth in the medium run following disaster exposure, but with a delayed effect. This is likely to be because of higher adjustment costs of capital investment.

Taken together, the dynamic results indicate that structured management practices are associated with smaller initial real effects of natural disasters on firms, particularly in terms of employment and survival. The resilience effect appears stronger in the initial year of disaster

exposure than in subsequent years. Nevertheless, we need to be cautious as across all flavours of dynamic models, the estimates become imprecise (especially at longer lags).

3.4 Heterogeneity and Robustness

In Appendix B.3, we conduct a battery of robustness checks of the results in Table 1 and analyze heterogeneity along several dimensions. We summarize some of these results here.

Considering first heterogeneity of the treatment effects, an important question is whether the protective effect of management varies with a country’s level of development. We examine this by defining a high-income country indicator and interacting it with all regressors in specification (3), allowing disaster effects, management effects, and their interaction to differ across income groups.¹⁵ Table B10 presents the results. In the survival regressions, the management–disaster interaction is positive, large and significant for the low-income base group, but the triple interaction with high income is significantly negative. This pattern is consistent with institutional substitution: in high-income countries, firms may have better access to insurance markets, disaster relief, and credit facilities that support survival regardless of internal management quality, whereas in low-income countries these external buffers are relatively absent and a firm’s own organizational capacity may be a more important margin for survival.

Second, we examine the degree of disaster severity, drawing on additional information recorded in EM-DAT regarding the number of recorded deaths and displaced population due to each disaster. We classify a disaster as ‘severe’ using two criteria. First, a disaster is deemed severe if it is in the top quartile of all disasters in terms of either deaths or number of displaced people (these thresholds correspond to 44 deaths and approximately 21,300 people displaced). Second, in addition to these aggregate thresholds, disasters in a particular country are categorized as severe if they are in the top 5% of disasters according to country-specific deaths or number of displaced people. Table B11 contains the results from including a severe disaster indicator alongside our baseline disaster measure. We find that the management results are concentrated among severe disasters, with the management-disaster interactions in the survival and employment growth regressions only significant for the severe disaster indicator.

Third, we disaggregate the overall management score. The overall management score is an average across 18 dimensions and it is conventional to group these dimensions into four categories of operations, monitoring, targets and people (HR practices) management. We re-estimate the baseline regressions including these four categories (again in z-scores) instead of the overall management score. The results in Table B12 lose some precision as using a smaller number of practices to construct the separate indices adds noise to the management variables, which are also strongly correlated with each other. Nonetheless, the targets dimension of management appears to have the strongest positive interaction. This captures the extent to which firms set ambitious (yet achievable) targets and regularly evaluate performance against them.

Next we consider robustness. First, as described in Section 2, our measure of disaster exposure is based on whether a firm is located within 40 km of a disaster recorded in GDIS.

15. We classify countries as high income according to their World Bank income level classification in 2018, which is the last year of our sample in this analysis (World Bank 2023). Countries’ income category is given in Table A1.

Figure B1 in Appendix B.3 shows that the estimated disaster–management interaction effects are broadly robust to alternative definitions of exposure using distance thresholds between 20 and 70 km. The interaction estimates are similar in magnitude, or larger, when using thresholds below 40 km, but attenuate as the distance threshold increases.

Second, some papers have found that decentralized firms are more resilient to negative shocks (e.g., Aghion et al. (2021)). The WMS survey also includes several questions on the degree of decentralization within a firm, reflecting the degree to which a plant manager has autonomy from the central headquarters. Since the structured management score is positively correlated with this measure, one may be concerned that our results are driven by decentralization rather than by management *per se*. Table B13 presents the results of panels D and E of Table 1 additionally controlling for decentralization and its interaction with disasters. Decentralized firms do have significantly higher employment growth and are more likely to survive. But, in contrast with management, none of the interactions between decentralization and disaster exposure are significant, whereas the coefficients on the disaster-management interactions in the employment growth and survival equations remain positive and significant

Third, while our baseline specification includes country, year and two-digit industry fixed effects, Table B14 additionally includes year dummies interacted with one-digit industry and year-by-region fixed effects. The estimates of the disaster-management interaction term on employment growth and on survival are similar to those presented in Table 1. Moreover, Table B15 shows that our results are robust to additionally including linear time trends that differ by two-digit industry and by country, while Table B16 shows they are robust to excluding the industry dummies.

Finally, columns (1)–(3) of Table 1 relate to firm performance conditional on survival. To examine whether selection among surviving firms is driving these estimates, we implement a bounds approach in the style of Lee (2009). To construct ‘lower bounds’ we replace missing employment growth for exiters with the bottom decile (p10) of employment growth and for an ‘upper bound’ we use the top decile (p90). The estimates were very similar to those in Table 1.¹⁶ The coefficients (standard errors) on the key interaction term between management and disaster in the estimates from Panel D were 0.0038(0.0014) and 0.0036(0.0014), very similar to the baseline.¹⁷

Overall, our results on the resilience effects of management for survival and job growth appear robust. In terms of heterogeneity, the greater resilience of firms with structured management practices seems particularly strong in developing countries, for the most severe disasters and for the targets element of management.

16. The coefficient (standard error) on the linear disaster effect in Panel D was -0.0027(0.0014) for the lower bound and -0.0022(0.0014) for the upper bound compared to -0.0024(0.0014) in the baseline.

17. We also considered a ‘survivor’ subsample, where we drop all observations from all firms who exited within one year of a disaster, with the aim of eliminating firms on negative pre-trends. Restricting the analysis to these surviving firms and those that never experience a disaster leads to no discernible change on the results.

4 Management Practices, Climate Risk Perceptions and Climate Change Adaptation

The results of the previous section suggest structured management practices protect firms from some of the negative impacts of natural disasters. There are numerous mechanisms through which this could arise. To structure our discussion of these mechanisms, it is useful to distinguish between pathways that operate via firms’ perceptions of disaster risk and pathways that operate independently from such perceptions.

Perception-driven mechanisms operate through firms’ awareness of climate-related risks. Within this class of mechanisms, we further distinguish between two distinct channels. The first, which we term ‘perception-contingent adaptation’, arises if management practices lead to more accurate perceptions of disaster risks in disaster-exposed locations, which in turn drive adaptive actions. This channel operates through the causal chain whereby management enhances risk awareness, which then motivates adaptation. The second perception-driven channel, which we term ‘enhanced perception-responsiveness’, operates differently. Even conditional on any given level of perceived risk, firms with more structured management may be more effective at translating their risk perceptions into concrete adaptive actions. This reflects superior organizational capacity to act on perceived threats, independent of whether management also improves the accuracy of those perceptions. The distinction between these two perception-driven mechanisms is important because they have different policy implications, which we discuss below.

Non-perception mechanisms, by contrast, can arise if firms with more structured management practices exhibit greater organizational resilience through capabilities that are in place independent of firms’ perceptions of disaster risk. Features such as internal reallocation capacity, supply-chain robustness, and access to financial liquidity can help mitigate the impact of a wide range of disruptions, including natural disasters. To the extent that these characteristics are more prevalent among firms with relatively structured management practices, they can insulate such firms from disaster impacts even if natural disasters are not regarded as a salient threat to their operations.

Beyond an analytic device, this three-way distinction is meaningful as the different mechanisms have different policy implications. If perception-contingent adaptation were the main reason firms with more structured management are resilient to disasters, then improving risk awareness in less structured firms may be sufficient to enhance their disaster resilience. Informational and educational campaigns on disasters are potential instruments. If enhanced perception-responsiveness were more important, then interventions would need to focus not just on risk awareness but also on organizational capabilities to translate awareness into action. If, by contrast, non-perception mechanisms were the dominant factor, then policy interventions targeting disaster risk awareness may be less effective than interventions that aimed to improve firms’ general management structures.

In this section, we provide evidence relevant to these mechanisms by exploiting questions in the latest wave of the WMS that asked firms about their perception of risks related to climate change and whether they have taken any actions to reduce these risk (i.e. whether they have en-

gaged in climate change adaptation).¹⁸ Evidence consistent with perception-driven adaptation would arise if structured management practices are positively associated with perceived climate risk or if the relationship between perceived risk and adaptation is stronger among firms with more structured management practices. A positive association between management and reported adaptation conditional on perceived disaster risk, meanwhile, would suggest that generic organizational resilience is an important mechanism.

We first examine evidence relevant to perception-driven mechanisms by analysing the association between management practices and perception of climate risks. Since we can only conduct this analysis on the most recent WMS wave, we consider different natural disaster exposure measures to those in the previous section.¹⁹ Rather than identify firms that have been exposed to a natural disaster in a particular year, we instead aggregate natural disasters by decade and construct binary indicators that take the value of 1 if a firm’s location is within 40km of at least one disaster in a particular decade and zero otherwise. These disaster exposure indicators can therefore be considered backward-looking measures of the riskiness of firms’ locations, in contrast to the measures in the previous section that captured whether a firm had itself been exposed to a disaster.

Because the wording of the climate risk perception and adaptation questions explicitly refers to *changes* in climate patterns, we focus on the *change* in disaster exposure over time rather than disaster exposure in any single decade. We do so by constructing $\Delta D_i = D_{i,2010s} - D_{i,1970s}$, where $D_{i,decade}$ is a binary indicator for whether firm i ’s location is within 40km of at least one disaster in the relevant decade. This variable takes the value 1 for firms in locations that were exposed to disasters in the 2010s but not in the 1970s (i.e. locations that have experienced an increase in disaster exposure), -1 for the reverse, and 0 for locations with no change.²⁰

We examine the impact of management practices on disaster perception using variants on the following specification:

$$P_i = \gamma_0 \Delta D_i + \gamma_1 M_i + \gamma_2 X_i + \gamma_3 (M_i * \Delta D_i) + \gamma_4 (X_i * \Delta D_i) + \iota_j + \kappa_c + \epsilon_i, \quad (5)$$

where P is a firm’s standardized risk perception score and $\Delta D_i = D_{i,2010s} - D_{i,1970s}$ measures the change in disaster exposure between the 1970s and the 2010s. We are specifically interested in the parameters γ_1 and γ_3 : γ_1 captures any overall association between structured management and climate risk perceptions, while γ_3 captures whether this association is stronger among firms in locations where disaster exposure has increased.

Table 2 examines how firms’ perceptions of climate change risk vary with changes in disaster exposure over recent decades and management practices. Column (1) shows that increases in disasters are positively, but insignificantly associated with risk perceptions. Column (2) shows

18. The specific wording of these questions is provided in Section 2.

19. Appendix Table B19 shows that the growth results of Table 1 are, if anything, somewhat stronger in the subsample of firms observed in the latest WMS wave, which we examine in this section. We are unable to estimate the survival regression on this subsample, since the data analyzed in Section 3 ends in 2018 whereas all firms observed in the latest WMS wave survive to at least 2021.

20. Tables B24 and B28 in the appendix show results of this section are robust to defining the ΔD_i variable as the difference between the *count* of disasters in the 2010s and the 1970s, rather than the difference in the binary indicator for any disaster in each decade. An earlier version of this paper used the two decade-specific $D_{i,2010s}$ and $D_{i,1970s}$ indicators rather than their difference, which yielded broadly similar results.

that firms with more structured management practices are significantly more likely to perceive climate risks compared to other firms. A one standard deviation in the management score is associated with an increase of 0.09 of a standard deviation in the risk perceptions index. The final column shows that the management coefficient is driven by firms located in areas that are experiencing an increase in disaster exposure relative to the 1970s.²¹

Table 2: Perceptions of Climate Change Risk, Disaster Exposure and Management

	(1)	(2)	(3)
ΔD_i	0.067 (0.067)	0.072 (0.065)	0.079 (0.068)
Management		0.093*** (0.035)	-0.001 (0.046)
(Management) * (ΔD_i)			0.212*** (0.068)
N obs.	1117	1117	1117
Industry Dummies	Yes	Yes	Yes
Country Dummies	Yes	Yes	Yes
Linear Firm Characteristics	No	Yes	Yes
Firm Char.-Disaster Interactions	No	No	Yes

Note: standard errors in parentheses clustered at the firm level. Dependent variable is a firm's standardized climate risk perception score. Firm characteristics include log(employment), firm age and share of the workforce with a degree. ΔD_i is the difference between two binary measures ($D_{i,2010s} - D_{i,1970s}$), where $D_{i,decade}$ indicates whether firm i 's location is within 40km of at least one disaster in the relevant decade. */**/** denote significance at the 10/5/1 percent level respectively.

To assess whether these patterns merely reflect generic responses to adverse shocks, rather than responsiveness to events with a clear link to climate change, we examine heterogeneity by disaster type and replicate the analysis separately for floods and storms, which are plausibly affected by climate change, and earthquakes, which are not. These results are reported in Appendix Tables B25 and B26. For floods and storms, the estimated effects closely mirror those obtained using the aggregate disaster measure. In contrast, for earthquakes there is no evidence that risk perceptions respond to changes in earthquake exposure either linearly or when interacted with management. The linear management coefficient remains positive and statistically significant in these specifications, indicating that firms with more structured management are, on average, more likely to perceive climate change as a risk, but do not appear to condition these perceptions on exposure to non-climate-related shocks.

Overall, Table 2 provides evidence that firms with more structured management processes are more likely to be aware of the risks posed by natural disasters than other firms (i.e. that they have more accurate disaster risk perceptions). This parallels the work of Bloom et al. (2025),

21. This pattern could be explained by at least two channels: firms with structured management practices may update their perceptions of climate risk to a greater extent following local disasters than other firms, or they may have strong general priors about the exposure of their local area to changing risk exposure. Since we only observe climate risk perceptions in the latest wave of the WMS, we are confined to cross-sectional analysis and therefore cannot distinguish between these alternative hypotheses.

who find a positive association between firms' management and the accuracy of their forecasts, both of their own year-ahead performance and that of the wider economy.²²

While Table 2 provides suggestive evidence that perception-driven adaptation may be a mechanism behind the greater resilience of some firms to natural disasters, greater perceptions of disaster risk alone are unlikely to materially influence resilience unless they prompt specific actions by firms to ready themselves for the realization of such risk. We therefore next examine climate adaptation directly by estimating variants of the following equation:

$$A_i = \delta_0 \Delta D_i + \delta_1 M_i + \delta_2 P_i + \delta_3 (M_i * P_i) + \delta_4 X_i + \delta_5 (M_i * \Delta D_i) + \delta_6 (X_i * \Delta D_i) + \iota_j + \kappa_c + \epsilon_i, \quad (6)$$

where A_i is firm i 's standardized climate adaptation score and all other notation is as before. In addition to firm attributes M_i and X_i , we also include firms' climate risk perception score P as an explanatory variable in some specifications, since it is plausible that firms who perceive climate change to be a risk to their business are doing more to reduce such risks. We also allow the propensity of firms to act on their risk perceptions to differ according to their management practices by including the interaction term $M_i * P_i$.

Table 3 contains estimates of the main parameters of interest from equation (6). Column (1) shows that the change in disaster exposure between the 1970s and the 2010s has a positive and significant association with climate risk adaptation. The management score has a large and strongly positive association with adaptation - a one standard deviation in the management score is associated with an increase of 0.31 of a standard deviation in the adaptation index. Column (2) includes the risk perception index which, as one might expect, enters with a positive and significant coefficient. The coefficient on management falls slightly, suggesting some of the adaptation-management relationship is through the fact that structured management aids greater risk perception as shown in Table 2. However, the fall of the coefficient on management is small, from 0.31 to 0.27. This implies that there are effects of management over and above any influence of perceptions. Column (3) includes an interaction between management and risk perception. The coefficient is large and significant, implying that firms with more structured management adapt more for a given level of risk perception. This is consistent with the enhanced perception-responsiveness hypothesis. Column (4) adds an interaction between management and the change in disaster exposure. The coefficient is small and insignificant, indicating that the strong direct association between management and adaptation documented in the preceding columns does not vary with local changes in disaster exposure. This is consistent with the management-adaptation relationship operating primarily through generic organizational capacity rather than through differential responses to local disaster trends. The final column shows that the results are robust to a wider set of controls.²³

22. Table B27 in Appendix B repeats the most comprehensive specification in Table 2 for the separate components of management practices. This shows the interaction between management and disaster exposure is positive and significant across all dimensions of management, with the strongest effect on the targets management score.

23. Tables B29 and B30 in Appendix B repeat the analysis of Table 3 separately for floods/storms and earthquakes. Both these types of natural disaster yield results similar to those of Table 3, although the linear effect of disaster exposure on adaptation is somewhat smaller for earthquakes than for floods/storms. Table B31 reports the most comprehensive specification in column (5) of Table 3 re-estimated using the different components of management. The linear management effect is strongest for the targets component, whereas the interaction

Table 3: Climate Change Adaptation, Disaster Exposure and Management

	(1)	(2)	(3)	(4)	(5)
ΔD_i	0.180*** (0.062)	0.159*** (0.061)	0.151** (0.061)	0.151** (0.061)	0.142** (0.061)
Management	0.310*** (0.032)	0.274*** (0.032)	0.267*** (0.032)	0.265*** (0.039)	0.266*** (0.040)
(Management) * (ΔD_i)				0.006 (0.058)	-0.007 (0.060)
Risk perception		0.216*** (0.030)	0.219*** (0.030)	0.219*** (0.030)	0.221*** (0.031)
(Management) * (Risk perception)			0.090*** (0.030)	0.090*** (0.030)	0.091*** (0.030)
N obs.	1117	1117	1117	1117	1117
Industry Dummies	Yes	Yes	Yes	Yes	Yes
Country Dummies	Yes	Yes	Yes	Yes	Yes
Linear Firm Characteristics	No	Yes	Yes	Yes	Yes
Firm Char.-Disaster Interactions	No	No	No	No	Yes

Note: standard errors in parentheses clustered at the firm level. Dependent variable is a firm's standardized climate change adaptation score. Firm characteristics include log(employment), firm age and share of the workforce with a degree. ΔD_i is the difference between two binary measures ($D_{i,2010s} - D_{i,1970s}$), where $D_{i,decade}$ indicates whether firm i 's location is within 40km of at least one disaster in the relevant decade. */**/** denote significance at the 10/5/1 percent level respectively.

We can also examine whether these mechanism variables have explanatory power in the main regressions, since we observe both the climate variables and firm outcomes for the subsample of firms surveyed in the latest WMS wave. Appendix Table B20 re-estimates Panels D and E of Table 1 on this subsample, adding the climate risk perception score and its interaction with disaster exposure. On the employment margin, the perception–disaster interaction is positive and significant (0.0057), and the management–disaster interaction attenuates slightly from 0.0089 to 0.0084. Appendix Table B21 repeats the exercise using the climate adaptation score in place of perceptions. The adaptation interaction is again positive and significant, indicating that firms reporting more adaptation measures experience smaller employment losses following a disaster, and the management interaction attenuates further, to 0.0070, consistent with adaptation capturing part of the channel through which management operates. Appendix Table B22 shows that when both climate variables are included simultaneously neither climate interaction remains individually significant, though the management–disaster interaction is essentially unchanged (0.0067). This likely reflects the combination of a small subsample and the positive correlation between the two variables (0.36 on the employment growth subsample), which together preclude separate identification of their effects. As in Table 1, we find no significant interactions on capital or value added growth.

The estimates in Tables B20, B21 and B22 should nonetheless be interpreted cautiously, between management and climate risk perceptions is strongest for the operations component.

since the climate variables are subject to greater reverse causality concerns than the management variable. A firm that suffered a disaster during the sample period may report greater adaptation and heightened risk perceptions in the survey precisely because of that experience, and the direction of causality between the climate scores and disaster impacts therefore cannot be established from these estimates. Repeated management scores across WMS waves allow us to test whether management responds to disaster exposure (Appendix B.2), but the climate variables are observed only once and no equivalent test is available. What the estimates do establish is that the firms whose perceptions and adaptation behavior we analyze in this section are the same firms exhibiting management-related resilience in Section 3, and that the climate variables carry information about disaster resilience related to, but not fully captured by, the management score.

Another consequence of the climate variables being measured at one point in time only is that the associations documented in Tables 2 and 3 are cross-sectional. Subject to these caveats, the results of this section nonetheless provide suggestive evidence that the greater disaster resilience of firms with more structured management, as documented in Section 3, operates through all three channels: perception-contingent adaptation (better risk awareness in relevant contexts leading to action), enhanced perception-responsiveness (superior ability to act on perceived risks) and non-perception mechanisms (generic organizational resilience). The presence of all three mechanisms suggests that policies to improve disaster resilience among firms will need to address both risk awareness and organizational capabilities, rather than focusing on either dimension in isolation.

5 Conclusions

Analysing data from 33 countries spanning 35 years, we find exposure to a natural disaster is associated with higher firm exit and lower growth in capital, employment and value added among surviving firms. More interestingly, we also find that structured management practices mitigate some of these negative effects, as firms with a higher management score experience smaller reductions in employment growth and lower exit rates following natural disaster exposure. Such greater resilience of firms with more structured management is primarily driven by differential performance following severe natural disasters and is strongest for firms' target setting practices. These results indicate that structured management practices are associated with smaller near-term real effects of disasters.

Looking beyond the immediate aftermath of a disaster, we find that the estimated interaction effects between management practices and disaster exposure remain generally positive at longer horizons, but become less precisely estimated. While these estimates are noisy, their pattern suggests attenuation rather than reversal: firms with less structured management tend to experience larger initial declines and subsequently recover, whereas more structured firms exhibit greater stability in the face of shocks. Together, the dynamic evidence points to management practices reducing the immediate severity of disaster impacts.

To understand the mechanisms behind these patterns, we distinguish between three channels through which management practices may enhance disaster resilience. Perception-contingent

adaptation occurs when management practices lead to more accurate risk perceptions in disaster-exposed locations which then motivate adaptive actions. Enhanced perception-responsiveness, by contrast, operates when firms with relatively structured management practices are more effective at translating any given level of risk perception into concrete measures. In addition to these two perception-based mechanisms, non-perception mechanisms operate through generic organizational capabilities, such as internal reallocation capacity, supply-chain robustness, or access to liquidity, which are in place independent of firms' awareness of disaster risks.

Drawing on novel survey data on climate risk perceptions and adaptation, we provide evidence in support of all three channels. We find that structured management practices are associated with differences in how firms process climate-relevant information: firms with more structured management located in areas where disasters have increased since the 1970s exhibit elevated risk perceptions compared to others, and are more effective at translating perceived risks into adaptive actions conditional on the level of those perceptions. At the same time, firms with more structured management report greater climate adaptation even conditional on their risk perceptions, suggesting that generic organizational capabilities also play an important protective role.

These findings have important but nuanced implications for policy. The presence of all three mechanisms suggests that policies to improve disaster resilience among firms with less structured management will need to address both risk awareness and organizational capabilities, rather than focusing on either dimension in isolation. Interventions aimed solely at increasing awareness of climate-related risks may increase firms' adoption of protective measures, but will have limited impact if organizational or financial constraints prevent firms from acting on heightened risk perceptions. Conversely, interventions designed to strengthen firms' organizational foundations — such as structured target-setting processes or practices that support supply chain flexibility — can enhance resilience without needing to influence firms' risk perceptions. The most effective policies are likely to combine elements that improve both risk awareness in relevant contexts and organizational capacity to act on that awareness.

At the same time, our findings also raise broader questions about the aggregate implications of firm-level resilience. While firms with more structured management reduce short-run disruption and exit risk, disaster-induced exit of firms with weaker management may potentially facilitate reallocation toward more efficient firms. Erda (2026) finds evidence of this, with exit of low-productivity firms following floods in the US prompting capital upgrading and reallocation that can lead to aggregate productivity gains after disasters. Whether policies should prioritize firm-level resilience or permit market-driven reallocation likely depends on the magnitude of adjustment costs and institutional context. In environments with high adjustment costs, such as rigid labor markets or limited access to finance for new entrants, policies that increase firms' structured management practices may avoid costly reallocation processes and preserve employment. In settings where reallocation is relatively frictionless, by contrast, selective exit following disasters may play a productive role in improving allocative efficiency. The appropriate policy response is therefore likely to be context-specific, depending on institutional features, the severity of transition costs, and social preferences regarding short-run stability versus long-run efficiency.

More generally, our analysis highlights the value of distinguishing between resilience driven by generic organizational capabilities and resilience that arises through more targeted adaptation. Understanding the relative importance of each of these mechanisms, how they interact with national institutions and market frictions, and how firm-level resilience shapes aggregate adjustment to climate change all represent important directions for future research. Integrating firm-level evidence on resilience with models of entry, exit, and reallocation would help clarify the aggregate consequences of climate-related shocks and the trade-offs implied by policies that target firm survival versus adjustment. Furthermore, examining how national institutions and policies such as labor market regulations or disaster relief programs shape the relationship between management practices and disaster responses would valuably inform the design of climate adaptation policy across different institutional contexts.

Our results also generate predictions that future work could test as appropriate data become available. The protective association of management in the employment and survival margins predicts that firms with more structured management retain workers through disruptions that lead others to shed them, which matched employer–employee data covering disaster-exposed areas could test directly. One possible reason such firms are able to do so is that they enter disasters with larger liquidity buffers or obtain finance more readily afterwards, which data on cash holdings and credit lines would help establish. The estimated interaction between management and risk perceptions in the adaptation equation, meanwhile, predicts that campaigns to raise climate risk awareness will yield larger increases in adaptation among firms that already have structured practices than among those that do not. This prediction is in principle testable in settings where informational interventions are randomized or quasi-randomly assigned across firms with varying management quality.

A large body of evidence shows that management practices are strongly associated with firm productivity and performance. Our findings indicate that their relevance extends to firms’ ability to cope with and adapt to natural disasters. As the economic consequences of climate change become more pronounced, understanding how managerial capabilities shape firms’ resilience, learning, and adjustment will be central to both managerial decision-making and policy design.

References

- Acevedo, Sebastian, Mico Mrkaic, Natalija Novta, Evgenia Pugacheva, and Petia Topalova. 2020. “The Effects of Weather Shocks on Economic Activity: What are the Channels of Impact?” *Journal of Macroeconomics* 65 (September): 103207.
- Akerberg, Daniel, Kevin Caves, and Garth Frazer. 2015. “Identification Properties of Recent Production Function Estimators.” *Econometrica* 83 (6): 2411–2451.
- Adhvaryu, Achyuta, Namrata Kala, and Anant Nyshadham. 2020. “The Light and the Heat: Productivity Co-Benefits of Energy-Saving Technology.” *The Review of Economics and Statistics* 102, no. 4 (October): 779–792.
- . 2022. “Management and Shocks to Worker Productivity.” *Journal of Political Economy* 130, no. 1 (January 1, 2022): 1–47.

- Aghion, Philippe, Nicholas Bloom, Brian Lucking, Raffaella Sadun, and John Van Reenen. 2021. “Turbulence, Firm Decentralization, and Growth in Bad Times.” *American Economic Journal: Applied Economics* 13, no. 1 (January 1, 2021): 133–169.
- Allcott, Hunt, Sendhil Mullainathan, and Dmitry Taubinsky. 2014. “Energy policy with externalities and internalities.” *Journal of Public Economics* 112 (April): 72–88.
- Auffhammer, Maximilian, and Tamma A. Carleton. 2018. “Regional Crop Diversity and Weather Shocks in India.” *Asian Development Review* 35, no. 2 (August): 113–130.
- Augier, Mie, and David J. Teece. 2009. “Dynamic Capabilities and the Role of Managers in Business Strategy and Economic Performance.” *Organization Science* 20, no. 2 (April): 410–421.
- Balboni, C, J Boehm, and M Waseem. 2023. “Firm Adaptation in Production Networks: Evidence from Extreme Weather Events in Pakistan.” *Working Paper*.
- Bandiera, Oriana, Andrea Prat, Stephen Hansen, and Raffaella Sadun. 2020. “CEO Behavior and Firm Performance.” *Journal of Political Economy* 128, no. 4 (April): 1325–1369.
- Barrot, Jean-Noël, and Julien Sauvagnat. 2016. “Input Specificity and the Propagation of Idiosyncratic Shocks in Production Networks *.” *The Quarterly Journal of Economics* 131, no. 3 (August 1, 2016): 1543–1592.
- Besley, Timothy, and Torsten Persson. 2009. “Repression or Civil War?” *American Economic Review* 99, no. 2 (April 1, 2009): 292–297.
- Bianchi, Nicola, and Michela Giorcelli. 2022. “The Dynamics and Spillovers of Management Interventions: Evidence from the Training within Industry Program.” *Journal of Political Economy* 130, no. 6 (June 1, 2022): 1630–1675.
- Bloom, N., and J. Van Reenen. 2007. “Measuring and Explaining Management Practices Across Firms and Countries.” *The Quarterly Journal of Economics* 122, no. 4 (November 1, 2007): 1351–1408.
- Bloom, Nicholas, Erik Brynjolfsson, Lucia Foster, Ron Jarmin, Megha Patnaik, Itay Saporta-Eksten, and John Van Reenen. 2019. “What Drives Differences in Management Practices?” *American Economic Review* 109, no. 5 (May 1, 2019): 1648–1683.
- Bloom, Nicholas, Christos Genakos, Ralf Martin, and Raffaella Sadun. 2010. “Modern Management: Good for the Environment or Just Hot Air?” *The Economic Journal* 120, no. 544 (May 1, 2010): 551–572.
- Bloom, Nicholas, Takafumi Kawakubo, Charlotte Meng, Paul Mizen, Rebecca Riley, Tatsuro Senga, and John Van Reenen. 2025. “Do Well Managed Firms Make Better Forecasts?” *The Review of Economics and Statistics*.
- Bloom, Nicholas, Renata Lemos, Olivia Marques, Raffaella Sadun, Daniela Scur, Marcos Todechini, and John Van Reenen. 2024. “Management matters in an era of disruption.” *POID Briefing Paper*.

- Bloom, Nicholas, Renata Lemos, Raffaella Sadun, Daniela Scur, and John Van Reenen. 2014. "The new empirical economics of management." *Journal of the European Economic Association* 12:835–76.
- Bloom, Nicholas, Aprajit Mahajan, David McKenzie, and John Roberts. 2020. "Do Management Interventions Last? Evidence from India." *American Economic Journal: Applied Economics* 12, no. 2 (April 1, 2020): 198–219.
- Borusyak, Kirill, and Peter Hull. 2023. "Nonrandom Exposure to Exogenous Shocks." *Econometrica* 91 (6): 2155–2185.
- Brakenridge, G.R. 2025. *Global Active Archive of Large Flood Events, 1985-Present*. Accessed June 4, 2025. <https://shorturl.at/1YeHo>.
- Bruhn, Miriam, Dean Karlan, and Antoinette Schoar. 2018. "The Impact of Consulting Services on Small and Medium Enterprises: Evidence from a Randomized Trial in Mexico." *Journal of Political Economy* 126, no. 2 (April): 635–687.
- Capelle, Damien, Divya Kirti, Nicola Pierri, and Germán Villegas Bauer. 2024. "Mitigating Climate Change at the Firm level: Mind the Laggards." *Working Paper*.
- Carvalho, Vasco M, Makoto Nirei, Yukiko U Saito, and Alireza Tahbaz-Salehi. 2021. "Supply Chain Disruptions: Evidence from the Great East Japan Earthquake*." *The Quarterly Journal of Economics* 136, no. 2 (March 24, 2021): 1255–1321.
- Cavallo, Eduardo, Sebastian Galiani, Ilan Noy, and Juan Pantano. 2013. "Catastrophic Natural Disasters and Economic Growth." *The Review of Economics and Statistics* 95, no. 5 (December 1, 2013): 1549–1561.
- Clò, Stefano, Francesco David, and Samuele Segoni. 2024. "The impact of hydrogeological events on firms: Evidence from Italy." *Journal of Environmental Economics and Management* 124 (March): 102942.
- Dar, Manzoor H., Alain de Janvry, Kyle Emerick, David Raitzer, and Elisabeth Sadoulet. 2013. "Flood-tolerant rice reduces yield variability and raises expected yield, differentially benefiting socially disadvantaged groups." *Scientific Reports* 3, no. 1 (November 22, 2013): 3315.
- Dell, Melissa, Benjamin F Jones, and Benjamin A Olken. 2012. "Temperature Shocks and Economic Growth: Evidence from the Last Half Century." *American Economic Journal: Macroeconomics* 4, no. 3 (July 1, 2012): 66–95.
- Elliott, Robert J.R., Eric Strobl, and Puyang Sun. 2015. "The local impact of typhoons on economic activity in China: A view from outer space." *Journal of Urban Economics* 88 (July): 50–66.
- Erda, T. 2026. "Cleansing Floods? Creative Destruction and Government Spending after Disasters." *Working Paper*.
- GADM. 2018. *Database of Global Administrative Areas*. <https://gadm.org/data.html>.

- Giorcelli, Michela. 2019. “The Long-Term Effects of Management and Technology Transfers.” *American Economic Review* 109, no. 1 (January 1, 2019): 121–152.
- Haas, R de, R Martin, M Muuls, and M Schweiger. 2022. “Managerial and financial barriers during the green transition.” *POID Working Paper* No.55.
- Huber, Kilian, Volker Lindenthal, and Fabian Waldinger. 2021. “Discrimination, Managers, and Firm Performance: Evidence from “Aryanizations” in Nazi Germany.” *Journal of Political Economy* 129, no. 9 (September 1, 2021): 2455–2503.
- Hultgren, Andrew, Tamma Carleton, Michael Delgado, Diana R. Gergel, Michael Greenstone, Trevor Houser, Solomon Hsiang, Amir Jina, Robert E. Kopp, Steven B. Malevich, Kelly McCusker, Terin Mayer, Ishan Nath, James Rising, Ashwin Rode, and Jiacan Yuan. 2022. “Estimating Global Impacts to Agriculture from Climate Change Accounting for Adaptation.” *SSRN Electronic Journal*.
- Iacovone, Leonardo, William Maloney, and David McKenzie. 2022. “Improving Management with Individual and Group-Based Consulting: Results from a Randomized Experiment in Colombia.” *The Review of Economic Studies* 89, no. 1 (January 10, 2022): 346–371.
- IPCC. 2023. *Intergovernmental Panel On Climate Change Climate Change 2022 – Impacts, Adaptation and Vulnerability: Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. 1st ed. Cambridge University Press, June 22, 2023. Accessed September 19, 2023. <https://www.cambridge.org/core/product/identifier/9781009325844/type/book>.
- Jones, Kyle, Wei Li, Paul Mizen, Rebecca Riley, and Jakob Schneebacher. 2026. *Managing to Adapt: Structured Management Practices and Firm Resilience*. IFS Working Paper W26/13. Institute for Fiscal Studies, February.
- Jordá, Óscar. 2005. “Estimation and Inference of Impulse Responses by Local Projections.” *American Economic Review* 95, no. 1 (March): 161–182.
- Kahn, Matthew E. 2005. “The Death Toll from Natural Disasters: The Role of Income, Geography, and Institutions.” *Review of Economics and Statistics* 87, no. 2 (May): 271–284.
- Kocornik-Mina, Adriana, Thomas K. J. McDermott, Guy Michaels, and Ferdinand Rauch. 2020. “Flooded Cities.” *American Economic Journal: Applied Economics* 12, no. 2 (April): 35–66.
- Lamorgese, Andrea, Megha Patnaik, Andrea Linarello, and Fabiano Schivardi. 2024. “Management Practices and Resilience to Shocks: Evidence from COVID-19.” *Management Science* 70 (12): 9058–9072.
- Lazear, Edward P., Kathryn L. Shaw, and Christopher T. Stanton. 2015. “The Value of Bosses.” *Journal of Labor Economics* 33, no. 4 (October): 823–861.
- Lee, David. 2009. “Training, Wages, and Sample Selection: Estimating Sharp Bounds on Treatment Effects.” *Review of Economic Studies* 76 (3): 1071–1102.

- Martin, Ralf, Mirabelle Muûls, Laure B. De Preux, and Ulrich J. Wagner. 2012. "Anatomy of a paradox: Management practices, organizational structure and energy efficiency." *Journal of Environmental Economics and Management* 63, no. 2 (March): 208–223.
- Masuda, Yuta J., Teevrat Garg, Ike Anggraeni, Kristie Ebi, Jennifer Krenz, Edward T. Game, Nicholas H. Wolff, and June T. Spector. 2021. "Warming from tropical deforestation reduces worker productivity in rural communities." *Nature Communications* 12, no. 1 (March 11, 2021): 1601.
- Michler, Jeffrey D., Kathy Baylis, Mary Arends-Kuenning, and Kizito Mazvimavi. 2019. "Conservation agriculture and climate resilience." *Journal of Environmental Economics and Management* 93 (January): 148–169.
- Rosvold, Elisabeth L., and Halvard Buhaug. 2021. "GDIS, a global dataset of geocoded disaster locations." *Scientific Data* 8, no. 1 (February 16, 2021): 61.
- Scur, Daniela, Raffaella Sadun, John Van Reenen, Renata Lemos, and Nicholas Bloom. 2021. "The World Management Survey at 18: lessons and the way forward." *Oxford Review of Economic Policy* 37, no. 2 (June 29, 2021): 231–258.
- Shin, Jieun. 2023. "Fuel Prices and Carbon Emissions: The Role of Structured Management Practices." *Working Paper*.
- Somanathan, E., Rohini Somanathan, Anant Sudarshan, and Meenu Tewari. 2021. "The Impact of Temperature on Productivity and Labor Supply: Evidence from Indian Manufacturing." *Journal of Political Economy* 129, no. 6 (June 1, 2021): 1797–1827.
- Tanaka, Ayumu. 2015. "The impacts of natural disasters on plants' growth: Evidence from the Great Hanshin-Awaji (Kobe) earthquake." *Regional Science and Urban Economics* 50 (January 1, 2015): 31–41.
- Teece, David J. 2007. "Explicating dynamic capabilities: the nature and microfoundations of (sustainable) enterprise performance." *Strategic Management Journal* 28, no. 13 (December): 1319–1350.
- World Bank. 2023. *World Bank Country and Lending Groups*. <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519>. Accessed September 2025.
- Zhang, Peng, Olivier Deschenes, Kyle Meng, and Junjie Zhang. 2018. "Temperature effects on productivity and factor reallocation: Evidence from a half million chinese manufacturing plants." *Journal of Environmental Economics and Management* 88 (March): 1–17.